MILITARY SPECIFICATION

CATAPULT, AIRCRAFT EJECTION SEAT, M3A1 METAL PARTS ASSEMBLY

1. SCOPE

1.1 This is the applicable detail specification for Catapult, Aircraft Ejection Seat, M3A1, Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 — Ammunition and Special Weapons, General Specification for.

MIL-I-45208 — Inspection Requirements, General Specification for.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Quality Assurance Terms and Definitions.

MIL-STD-413 — Visual Inspection Guide for Rubber O-Rings.

MIL-STD-639 — Surface Roughness Values Preferred for Army Ordnance Materiel.

DRAWINGS

MUNITIONS COMMAND

F8593636 — Catapult, Aircraft Ejection Seat, M3A1, Metal Parts Assembly.

F8797431 — Ammunition Container, Fiber M304 for Catapult, Aircraft Ejection Seat, M3A1.

F8797432 — Box, Packing, Ammunition for Catapult, Aircraft Ejection Seat, M3A1.

FSC 1375

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IEL–8593636 — Index to Inspection Equipment List for Catapult, Aircraft Ejection Seat, M3A1, Metal Parts Assembly.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Initial production sample. Unless otherwise directed by the contracting officer, an initial production sample shall be required.

3.3 Material. The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

3.4 Physical properties. Physical properties required by the drawing shall be uniform throughout each part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.5 Hydrostatic pressure. The components listed below shall withstand the corresponding internal pressure as indicated in table I for a period of 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure (see 6.1).

TABLE I

Component	Hydrostatic pressure (psi minimum)
Outside Tube	2500
Telescoping Tube	3200
Inside Tube	3200

3.6 Shear pin strength. After application of protective finish, the pin shall shear when a load of 400 plus or minus 50 pounds is applied at the steady rate of .25 inches per minute.

3.7 Structural strength. The metal parts assembly shall be capable of withstanding each test load for 15 seconds, minimum, as indicated in table II. The maximum allowable change between mounting points, resulting from each load application shall not exceed the values indicated in table II. A preload of 500 pounds shall be utilized prior to each test load application.

TABLE II

Test loads (including pre-load) min.	Change between mounting points, max.
1. 2000 pounds in tension	.005 inches
2. 2000 pounds in tension	.001 inches
3. 8000 pounds in compression	.005 inches
4. 8000 pounds in compression	.001 inches

3.8 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing and the following:

3.8.1 *Torque.* Torque requirements shall be as specified in table III. No loosening or deformation shall occur upon application of the breakaway torque.

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TABLE III

Component	A Run-on (inch pounds)	B Seating or bottoming (including A) (inch pounds)	C Breakaway (inch pounds)
Cap and Outside Tube	75 min.	550 + 50	400 min.
Trunnion and Outside Tube	75 min.	550 + 50	400 min.
Spacer and Telescoping Tube	.....	.....	200 min.
Block and Inside Tube	75 min.	.....	.....
Stop Ring and Trunnion	75 min.	.....	.....

3.8.1.1 *Bottoming torque.* The bottoming torque between block and inside tube and between stop ring and trunnion shall not exceed the running torque by more than 50 inch pounds.

3.9 Protective finish. Protective finish shall be in accordance with the applicable specification and drawings and shall apply to all surfaces of the component regardless of contour configuration.

3.10 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.11 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.11.1 *Metal defects.* All components shall be free from cracks, splits, cold shuts, inclusions, porosity or any similar defect.

3.11.2 *Bur.* No part shall have a bur which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.11.3 *Foreign matter.* No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Quality assurance terms and definitions.* Reference shall be made to MIL–STD–109 to define the quality assurance terms used.

4.1.2 *Inspection.* Inspection shall be in accordance with MIL–A–2550 and MIL–I–45208, except as specified herein.

4.2 Initial production sample. Prior to the initiation of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of 5 metal parts assemblies, 10 sets of unassembled parts and fifty shear pins from the same lot of shear pins tested by the contractor. The samples shall be subjected to the tests in 4.3.3. Test procedures shall be in accordance with 4.4, except as provided in MIL–I–45208. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing,

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must be the same as will be used in regular production and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system and other inspection requirements of MIL–I–45208. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved facility for all the requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted.

4.3 Inspection provisions.

4.3.1 *Lot.*

4.3.1.1 *Presentation of lots.* The product shall be presented in accordance with MIL–STD–105.

4.3.1.2 *Size of lot.* The shipping lots of metal parts assemblies shall consist of 621 metal parts assemblies, maximum, provided the homogeneity requirements of inspection lots have been retained. A smaller lot may be shipped upon approval of the contracting officer.

4.3.2 *Examination.* One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in tables IV and V. For an individual defect (single characteristic) the criteria in table IV shall be used. For total defects within a major or minor group the criteria in table V shall be used. All nonconforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL–STD–413. The Ordnance Standard Finish Blocks prescribed in MIL–STD–639 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE IV

Lot size	Sample size	Major		Minor	
		Accept	Reject	Accept	Reject
1 – 300	100% Examination				
301 – 1,300	110	1	2	2	3
1,301 – 3,200	150	2	3	3	4
3,201 – 8,000	225	3	4	4	5
8,001 – 22,000	300	4	5	5	6

TABLE V

	Total allowable defects per sample size			
Sample sizes for major defects	110	150	225	300
No. of Defects in Class				
2 – 3	1	2	3	4
4 – 7	2	3	4	5
8 – 19	3	4	5	7
20 – 43	4	5	8	10
over 43	6	8	11	14
2 – 3	2	3	4	5
4 – 7	3	4	5	7
8 – 16	4	5	8	10
17 – 48	6	8	11	14
over 48	8	11	17	20

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4.3.2.1 *Tube, Outside, Dwg. D8593638, covering a detail of Dwg. F8593636.*

Defect	*Method of inspection*
Critical	
1. Metal defective	Visual
Major	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of pitch diameter of thread with inside datum diameter	Gage
104. Concentricity of pitch diameter of thread with outside datum diameter.	Gage
105 Diameter of body, min. (within specified limit)	Gage
106. Diameter of body, min.	Gage
107. Diameter inside, max.	Gage
108. Total length, min.	Gage
109. Thickness of wall, min.	Gage
110. Straightness	Gage
111. Length of effective thread, min.	Measure
112. Length of thread, max.	Measure
113. Thread damaged or not full	Visual
114. Protective finish inadequate or defective	Visual
115. Surface finish, improper	Visual
116. Bur	Visual
117. Foreign matter	Visual
Minor	
201. Pitch diameter of thread, max.	Gage
202. Diameter of body, max. (within specified limit)	Gage
203. Diameter of body, max.	Gage
204. Diameter inside, min.	Gage
205. Total length, max.	Gage
206. Thread chamfer missing or incorrect	Visual
207. Marking missing, incorrect or illegible	Visual

4.3.2.2 *Tube, Outside and Pellet Assembly, Dwg. C8593639, covering a detail of Dwg. F8593636.*

Defect	*Method of inspection*
Critical	
None defined.	
Major	
101. Thread damaged	Visual
102. Bur	Visual
103. Foreign matter	Visual
Minor	
201. Diameter over top of pellet	Gage
202. Location of pellet	Gage
203. Angular location of pellet	Gage
204. Pellet missing, loose or improperly inserted	Manual/Visual
205. Pellet radius, incorrect	Visual

4.3.2.3 *Tube, Telescoping, Dwg. D8593640, covering a detail of Dwg. F8593636.*

Defect	*Method of inspection*
Critical	
1. Diameter of body, max.	Gage

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Defect	Method of inspection
2. Holes in large shoulder missing	Visual
3. Metal defective	Visual
Major	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of pitch diameter of thread with body at datum location	Gage
104. Diameter of body, min.	Gage
105. Diameter of shoulder, min.	Gage
106. Diameter inside, max.	Gage
107. Diameter of body adjacent to shoulder	Gage
108. Diameter of thread undercut, min.	Gage
109. Total length, max.	Gage
110. Wall thickness variation	Gage
111. Straightness	Gage
112. Thread damaged or not full	Visual
113. Number of holes in shoulder, incorrect	Visual
114. Protective finish inadequate or defective	Visual
115. Surface finish, improper	Visual
116. Bur	Visual
117. Foreign matter	Visual
Minor	
201. Pitch diameter of thread, max.	Gage
202. Diameter of shoulder, max.	Gage
203. Diameter inside, min.	Gage
204. Total length, min.	Gage
205. Diameter of holes in shoulder	Measure
206. Length of thread including undercut	Measure
207. Length to shoulder	Measure
208. Length of body adjacent to shoulder	Measure
209. Width of thread undercut, max.	Measure
210. Location of holes in shoulder	Measure
211. Marking missing, incorrect or illeglible	Visual

4.3.2.4 *Ring, Stop, Dwg. C8593642, covering a detail of Dwg. F8593636.*

Defect	Method of inspection
Critical	
1. Diameter of bore, min.	Gage
Major	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of bore with pitch diameter of thread	Gage
104. Concentricity of counterbore with bore	Gage
105. Symmetry of width across flats with flange	Gage
106. Diameter of thread undercut, min.	Gage
107. Length to counterbore	Gage
108. Length to datum diameter	Gage
109. Length to flange	Gage
110. Angle at datum diameter	Gage
111. Thread damaged or not full	Visual
112. Metal defective	Visual
113. Protective finish inadequate or defective	Visual
114. Surface finish of bore and chamfer on bore, improper	Visual
115. Bur or sharp corner	Visual
116. Foreign matter	Visual

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Defect	Method of inspection
Minor	
201. Pitch diameter of thread, max.	Gage
202. Diameter of bore, max.	Gage
203. Diameter of counterbore	Gage
204. Diameter of flange	Gage
205. Total length	Gage
206. Width across flats	Gage
207 Width of thread undercut, max.	Measure
208. Marking missing, incorrect or illegible	Visual
209. Radius or chamfer, missing or incorrect	Visual
210. Surface finish, improper, except as otherwise classified	Visual

4.3.2.5 *Ring, Stop, and Pellet Assembly, Dwg. C8593643, covering a detail of Dwg. F8593636.*

Defect	Method of inspection
Critical	
None defined.	
Major	
101. Thread damaged	Visual
102. Bur	Visual
103. Foreign matter	Visual
Minor	
201. Location of pellet	Gage
202. Angular location of pellet	Gage
203. Protrusion of pellet over thread	Gage
204. Pellet missing, loose or improperly inserted	Manual/Visual
205. Pellet radius, incorret	Visual

4.3.2.6 *Latch, Dwg. C8593644, covering a detail of Dwg. F8593636.*

Defect	Method of inspection
Critical	
1. Metal defective	Visual
Major	
101. Diameter of firing pin relief	Gage
102. Total length	Gage
103. Length to inside surface at small end	Gage
104. Length of cutout	Gage
105. Width at small end	Gage
106. Width at cutout	Gage
107. Thickness at datum locations	Gage
108. Location of firing pin relief	Gage
109. Angle at large end	Gage
110 . Protective finish inadequate or defective	Visual
111. Surface finish, improper	Visual
112. Bur or sharp corner	Visual
113. Foreign matter	Visual
Minor	
201. Thickness	Gage
202. Thickness at firing pin relief	Gage
203. Width at large end	Gage
204. Profile of large end	Gage
205. Profile of sides	Measure
206. Chamfer on small end at cutout	Measure
207. Marking missing, incorrect or illegible	Visual

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Defect	Method of inspection
Minor	
208. Radius or chamfer, missing or incorrect, except as otherwise classified	Visual

4.3.2.7 *Tube, Inside, Dwg. D8593646, covering a detail of Dwg. F8593636.*

Defect	Method of inspection
Critical	
1. Metal defective	Visual
Major	
101. Pitch diameter of thread, min.	Gage
102. Major diameter of thread, min.	Gage
103. Concentricity of pitch diameter of thread with outside diameter at datum location	Gage
104. Concentricity of pitch diameter of thread with inside diameter at datum location	Gage
105. Diameter of body	Gage
106. Diameter of body, midsection, min.	Gage
107. Diameter inside, max.	Gage
108. Depth of spiral groove	Gage
109. Depth of relief of spiral groove	Gage
110. Length of effective thread, min.	Gage
111. Total length, max.	Gage
112. Thickness of wall, min.	Gage
113. Straightness	Gage
114. Threads damaged or not full	Visual
115. Protective finish inadequate or defective	Visual
116. Surface finish of outside diameter of body at ends	Visual
117. Bur	Visual
118. Foreign matter	Visual
Minor	
201. Pitch diameter of thread, max.	Gage
202. Diameter inside, min.	Gage
203. Diameter of body midsection, max.	Gage
204. Total length, min.	Gage
205. Lenth of shoulder	Measure
206. Length of relief of spiral groove, max.	Measure
207. Length of thread, max.	Measure
208. Width of relief of spiral groove, max.	Measure
209. Location of spiral groove	Measure
210. Profile of spiral groove	Visual
211. Direction of turn of spiral groove	Visual
212. Number of turns of spiral groove, incorrect	Visual
213. Marking missing, incorrect or illegible	Visual
214. Radius or chamfer, missing or incorrect	Visual
215. Surface finish, improper, except as otherwise classified	Visual

4.3.2.8 *Tube, Inside, and Pellet Assembly, Dwg. C8593647, covering a detail of Dwg. F8593636.*

Defect	Method of inspection
Critical	
None defined.	
Major	
101. Thread damaged	Visual
102. Bur	Visual
103. Foreign matter	Visual

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Defect	Method of inspection
Minor	
201. Location of pellet	Gage
202. Angular location of pellet	Gage
203. Protrusion of pellet overthread	Gage
204. Pellet missing, loose or improperly inserted	Manual/Visual
205. Pellet radius, incorrect	Visual

4.3.2.9 *Block, Dwg. F8593648, covering a detail of Dwg. F8593636.*

Defect	Method of inspection
Critical	
None defined.	
Major	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Concentricity of counterbore with small cavity	Gage
104. Concentricity of pitch diameter of thread with counterbore and body	Gage
105. Concentricity of large cavity with counterbore	Gage
106. Concentricity of body shoulder with body	Gage
107. Concentricity of flange with body shoulder	Gage
108. Symmetry of length of slot with body shoulder	Gage
109. Symmetry of thickness of rectangular section with body shoulder	Gage
110. True position of hole in rectangular section with shoulder including perpendicularity with end of mounting lug	Gage
111. Symmetry of width of rectangular section with body shoulder	Gage
112. True position of screw hole with counterbore and base of large cavity.	Gage
113. True position of pin hole with small cavity	Gage
114. Diameter of small cavity	Gage
115. Diameter of counterbore, max.	Gage
116. Diameter of hole in rectangular section	Gage
117. Depth of large cavity	Gage
118. Length from flange to base of large cavity	Gage
119. Length to flange, max.	Gage
120. Length of effective thread, min.	Gage
121. Width of rectangular section	Gage
122. Thickness of rectangular section	Gage
123. Location of slot to base of large cavity	Gage
124. Diameter of thread undercut, max., if applicable	Measure
125. Profile of end of rectangular section	Measure
126. Threads damaged or not full	Visual
127. Metal defective	Visual
128. Protective finish inadequate or defective	Visual
129. Surface finish of small cavity, body shoulder or body slots, improper.	Visual
130. Bur	Visual
131. Foreign matter	Visual
Minor	
201. Pitch diameter of thread, min.	Gage
202. Pitch diameter of pipe thread	Gage
203. Diameter of counterbore, min.	Gage
204. Diameter of large cavity	Gage
205. Diameter of body	Gage
206. Diameter of body shoulder	Gage
207. Diameter of flange	Gage
208. Diameter of pin hole in body	Gage
209. Diameter of screw hole	Gage
210. Depth to bottom of small cavity	Gage

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Defect	Method of inspection
Minor	
211. Length to flange, min.	Gage
212. Length to end of flange	Gage
213. Length of slot in body	Gage
214. Total length	Gage
215. Width of slot in body	Gage
216. Location of hole in rectangular section	Gage
217. Location of screw hole and angular location with slot	Gage
218. Location of pin hole in body and angular location with slot	Gage
219. Diameter of undercut on body shoulder, min.	Measure
220. Diameter of drilled hole at bottom of small cavity, min.	Measure
221. Depth of counterbore, min.	Measure
222. Depth to bottom of drilled hole, max.	Measure
223. Length of chamfer at small cavity	Measure
224. Length of shoulder to bottom of flange	Measure
225. Width of undercut on body shoulder, max.	Measure
226. Location of threaded pipe hole	Visual
227. Marking missing or incorrect	Visual
228. Radius or chamfer, missing or incorrect	Visual
229. Surface finish, improper, except as otherwise classified	Visual

4.3.2.10 *Trunnion, Dwg. D8593649, covering a detail of Dwg. F8593636.*

Defect	Method of inspection
Critical	
None defined.	
Major	
101. Pitch diameter of small thread, max.	Gage
102. Minor diameter of small thread, max.	Gage
103. Pitch diameter of large thread, max.	Gage
104. Minor diameter of large thread, max.	Gage
105. Concentricity of small cavity with pitch diameter of small thread	Gage
106. Concentricity of pitch diameter of large thread with small cavity	Gage
107. Concentricity of large cavity with pitch diameter of large thread	Gage
108. Concentricity of lugs	Gage
109. Symmetry of width across flats at root of lugs with small cavity	Gage
110. True position of lug with small cavity	Gage
111. Diameter of small cavity	Gage
112. Diameter of large cavity	Gage
113. Diameter of lugs	Gage
114. Diameter of small thread undercut, max.	Gage
115. Diameter of large thread undercut, max.	Gage
116. Depth of small thread cavity, min.	Gage
117. Depth of spiral groove	Gage
118. Length of effective thread, min.	Gage
119. Width across flats at root of lugs	Gage
120. Thread damaged or not full	Visual
121. Metal defective	Visual
122. Protective finish inadequate or defective	Visual
123. Surface finish of small cavity, improper	Visual
124. Bur or sharp corner	Visual
125. Foreign matter	Visual
Minor	
201. Pitch diameter of small thread, min.	Gage
202. Pitch diameter of large thread, min.	Gage
203. Diameter of small end	Gage
204. Diameter of groove, small end	Gage

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Defect	Method of inspection
Minor	
205. Depth of small thread cavity, max.	Gage
206. Depth of relief of spiral groove	Gage
207. Length from base of large cavity to small end	Gage
208. Total length	Gage
209. Width of groove, small end	Gage
210. Location of groove, small end	Gage
211. Location of lug	Gage
212. Diameter at large end	Measure
213. Diameter of flange	Measure
214. Length of large thread	Measure
215. Length of flange	Measure
216. Location of spiral groove	Measure
217. Height of lug	Measure
218. Diameter of tang hole at spiral groove	Visual
219. Symmetry of tang hole with spiral groove	Visual
220. Width of large thread undercut, max.	Visual
221. Direction of spiral groove, incorrect	Visual
222. Number of turns of spiral groove, incorrect	Visual
223. Profile of spiral groove, incorrect	Visual
224. Relief of spiral groove, incorrect	Visual
225. Marking missing, incorrect or illegible	Visual
226. Radius, chamfer or both, missing or incorrect	Visual
227. Surface finish, improper, except as otherwise classified	Visual

4.3.2.11 *Cap, Dwg. C8593650, covering a detail of Dwg. F8593636.*

Defect	Method of inspection
Critical	
None defined.	
Major	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Concentricity of outside diameter with pitch diameter of thread	Gage
104. Diameter of thread undercut, max.	Gage
105. Diameter of body, min.	Gage
106. Thickness of base	Gage
107. Depth of base hole, max.	Measure
108. Width of thread undercut, max.	Measure
109. Thread damaged or not full	Visual
110. Metal defective	Visual
111. Protective finish inadequate or defective	Visual
112. Bur or sharp corner	Visual
113. Foreign matter	Visual
Minor	
201. Pitch diameter of thread, min.	Gage
202. Diameter of body, max.	Gage
203. Diameter of base hole	Measure
204. Depth of base hole, min.	Measure
205. Length between base holes	Measure
206. Total length	Measure
207. Location of base hole	Measure
208. Marking missing, incorrect or illegible	Visual
209. Radium or chamfer, missing or incorrect	Visual
210. Surface finish, improper	Visual

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4.3.2.12 *Spacer, Dwg. C8593651, covering a detail of Dwg. F8593636.*

Defect	Method of inspection
Critical	
None defined.	
Major	
101. Pitch diameter of thread, max.	Gage
102. Minor diameter of thread, max.	Gage
103. Concentricity of outside diameter with pitch diameter of threads	Gage
104. True position of holes with pitch diameter of thread	Gage
105. Number of holes incorrect	Visual
106. Threads damaged or not full	Visual
107. Protective finish inadequate or defective	Visual
108. Surface finish on outside diameter, improper	Visual
109. Bur or sharp edge	Visual
110. Foreign matter	
Minor	
201. Pitch diameter of thread, min.	Gage
202. Outside diameter	Gage
203. Thickness	Measure
204. Diameter of holes	Visual
205. Location of holes	Visual
206. Marking missing, incorrect or illegible	Visual
207. Radius or chamfer, missing or incorrect	Visual
208. Surface finish, improper, except as otherwise classified	Visual

4.3.2.13 *Washer, Shock, Dwg. B8593652, covering a detail of Dwg. F8593636.*

Defect	Method of inspection
Critical	
None defined.	
Major	
101. Tear, crack or split	Visual
Minor	
201. Thickness	Measure
202. Flash	Visual
203. Foreign matter	Visual

4.3.2.14 *Pin, Shear, Dwg. B8593654, covering a detail of Dwg. F8593636.*

Defect	Method of inspection
Critical	
None defined.	
Major	
101. Diameter, max.	Gage
102. Total length, min.	Gage
103. Metal defective	Visual
104. Protective finish, inadequate or defective	Visual
105. Surface finish, improper	Visual
106. Bur	Visual
107. Foreign matter	Visual
Minor	
201. Total length, max.	Gage
202. Profile of end	Measure

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4.3.2.15 *Plug, Dwg. C8593656, covering a detail of Dwg. F8593636.*

	Defect	*Method of inspection*
Critical		
1.	Diameter of boss hole, min.	Gage
Major		
101.	*Concentricity of boss hole with cavity*	Gage
102.	Concentricity of flange with cavity	Gage
103.	Concentricity of body with flange	Gage
104.	True position of thread with flange including perpendicularity with slot	Gage
105.	Symmetry of slot with cavity	Gage
106.	True position of pin hole with cavity and slot	Gage
107.	Diameter of cavity	Gage
108.	Diameter of flange, min.	Gage
109.	Depth of cavity, min.	Gage
110.	Depth of slot	Gage
111.	Total length, max.	Gage
112.	Height of boss	Gage
113.	Metal defective	Visual
114.	Protective finish inadequate or defective	Visual
115.	Surface finish of cavity and slot, improper	Visual
116.	Bur or sharp corner	Visual
117.	Foreign matter	Visual
Minor		
201.	Pitch diameter of thread	Gage
202.	Minor diameter of thread, max.	Gage
203.	Diameter of boss hole, max.	Gage
204.	Diameter of flange, max.	Gage
205.	Diameter of body	Gage
206.	Diameter of pin hole	Gage
207.	Depth of cavity, max.	Gage
208.	Depth of threaded hole, max.	Gage
209.	Total length, min.	Gage
210.	Length from slotted end to flange	Gage
211.	Length of effective thread, min.	Gage
212.	Width of slot	Gage
213.	Location of pin hole	Gage
214.	Location of threaded hole and angular location with slot	Gage
215.	Diameter of boss	Measure
216.	Diameter of undercut, min.	Measure
217.	Width of undercut, max.	Measure
218.	Angle of boss	Measure
219.	Chamfer on boss hole	Measure
220.	Marking missing, incorrect or illegible	Visual
221.	Radius or chamfer, missing or incorrect	Visual
222.	Surface finish, improper, except as otherwise classified	Visual

4.3.2.16 *Pin, Firing, Dwg. C8593657, covering a detail of Dwg. F8593636.*

	Defect	*Method of inspection*
Critical		
1.	Diameter of striker, max.	Gage
Major		
101.	Concentricity of striker with striker shoulder	Gage
102.	Concentricity of flange with striker shoulder	Gage
103.	Concentricity of groove with flange	Gage

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Defect	Method of inspection
Major	
104. Symmetry of forward datum location with striker shoulder	Gage
105. Symmetry of width of body with striker shoulder	Gage
106. Symmetry of thickness of body with striker shoulder	Gage
107. True position of pin hole with striker shoulder and body thickness	Gage
108. Diameter of striker shoulder, min.	Gage
109. Diameter of flange, min.	Gage
110. Diameter of groove	Gage
111. Depth of hole in striker shoulder, max.	Gage
112. Length of striker	Gage
113. Width at forward datum location	Gage
114. Width at rear datum location	Gage
115. Width of body	Gage
116. Thickness of body	Gage
117. Profile of groove	Measure
118. Relief groove on striker shoulder missing	Visual
119. Metal defective	Visual
120. Protective finish inadequate or defective	Visual
121. Surface finish of body, flange and striker shoulder, improper	Visual
122. Bur	Visual
123. Foreign matter	Visual
Minor	
201. Diameter of striker, min.	Gage
202. Diameter of striker shoulder, max.	Gage
203. Diameter of flanges, max.	Gage
204. Diameter of hole in striker shoulder	Gage
205. Depth of hole in striker shoulder, min.	Gage
206. Total length	Gage
207. Length from rear face of flange to striker shoulder	Gage
208. Length across relief grooves of striker shoulder	Gage
209. Width across flanges	Gage
210. Location of hole	Gage
211. Surface variation on rear face of striker shoulder	Gage
212. Diameter of lathe center hole, max.	Measure
213. Width of rear flange	Measure
214. Width of groove	Measure
215. Depth of relief groove on striker shoulder, min.	Visual
216. Radius or chamfer, missing or incorrect	Visual
217. Sharp corner where required	Visual
218. Marking missing, incorrect or illegible	Visual

4.3.2.17 *Seal, Plain, Dwg. B8593658, covering a detail of Dwg. F8593636.*

Defect	Method of inspection
Critical	
None defined.	
Major	
101. Diameter of wire, max.	Measure
102. Outside diameter, min.	Measure
103. Locational variance of ends, max.	Visual
104. Number of coils incorrect	Visual
105. Metal defective	Visual
106. Protective finish inadequate or defective	Visual
107. Bur or sharp edge	Visual
108. Foreign matter	Visual

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Defect	Method of inspection
Minor	
201. Outside diameter, max.	Measure
202. Length between coils	Measure
203. Direction of turn incorrect	Visual
204. Radius or chamfer, missing or incorrect	Visual

4.3.2.18 *Seal, Fixed, Dwg. B8593660, covering a detail of Dwg. F8593636.*

Defect	Method of inspection
Critical	
None defined.	
Major	
101. Diameter of wire, max.	Measure
102. Outside diameter, max.	Measure
103. Length of tang	Measure
104. Locational variance of ends, max.	Visual
105. Number of coils, incorrect	Visual
106. Metal defective	Visual
107. Protective finish inadequate or defective	Visual
108. Bur or sharp edge	Visual
109. Foreign matter	Visual
Minor	
201. Outside diameter, min.	Measure
202. Length between coils	Measure
203. Radius of tang bend	Measure
204. Direction of turn, incorrect	Visual
205. Radius or chamfer, missing or incorrect	Visual

4.3.2.19 *Metal Parts Assembly, Dwg. F8593636.*

Defect	Method of inspection
Critical	
None defined.	
Major	
101. Any part missing, except as otherwise classified	Manual/Visual
102. Improper assembly	Visual
103. Thread damaged or not full	Visual
104. Marking missing, incorrect or illegilbe	Visual
105. Bur	Visual
106. Foreign matter	Visual
107 Spacer not properly positioned	Measure
Minor	
201. Shipping plug missing	Visual
202. Location of ordnance part number on outside, tube, incorrect	Visual
203. Protective finish, inadequate or defective	Visual

4.3.2.20 *Fiber Container, Dwg. F8797431.*

Defect	Method of inspection
Critical	
None defined.	
Major	
None defined.	
Minor	
201. Contents loose	Manual

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Defect	Method of inspection
Minor	
202. Metal ends loose or distorted	Manual
203. Gap between body and cover, max.	Visual
204. Cut or damaged through all impregnated layers	Visual

4.3.2.21 *Packing Box, Dwg. F8797432.*

Defect	Method of inspection
Critical	
None defined.	
Major	
None defined.	
Minor	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Visual/Manual
203. Strapping missing, loose, broken or improperly assembled	Visual/Manual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual

4.3.2.22 *Sealing and retaining—compound application.* Each telescoping tube and spacer assembly shall be examined for presence of sealant on all surfaces other than the threaded joint of telescoping tube and spacer. Presence of sealant shall be cause for rejection of the assembly.

4.3.3 *Inspection testing.* The following tests shall be performed in accordance with the provisions of 4.4 except as provided in MIL–I–45208.

4.3.3.1 *Hardness.* Fifty units from each heat treated lot of components shall be tested to determine compliance with the drawing requirements. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.3.3.2 *Hydrostatic pressure.* Each outside, telescoping and inside tube shall be subjected to an internal hydrostatic pressure. Failure of any tube to comply with the requirements of 3.5 shall be cause for rejection of the tube.

4.3.3.3 *Shear pin strength.* Fifty pins from each lot shall be tested for shear strength. Failure of any pin to comply with the requirements of 3.6. shall be cause for rejection of the lot.

4.3.3.4 *Structural strength.* A sample of 15 complete assemblies shall be selected and subjected to the structural strength test. Failure of any assembly to comply with the requirements of 3.7 shall be cause for rejection of the lot.

4.3.3.5 *Breakaway torque.* A breakaway torque shall be applied to each assembly threaded joint as listed in table III. Failure of any assembly to comply with the requirements of column C of table III shall be cause for rejection of the assembly.

4.3.3.6 *Protective finish.* The number of specimens and acceptance criteria for protective finish tests shall be in accordance with the applicable Military Standard listed on the drawing. Failure of any sample to comply with the requirements of 3.9 or the applicable standard shall be cause for rejection of the lot.

4.3.3.7 *Residual magnetism.* Each metal parts assembly shall be tested for excessive residual magnetism. Failure of any assembly

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to comply with the requirements of 3.10 shall be cause for rejection of the assembly.

4.3.4 *Inspection equipment.* The examination and tests shall be made using equipment listed on IEL–8593636, except as provided in MIL–I–45208.

4.4 Test methods and procedures.

4.4.1 *Hardness.* Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the hardness requirements specified on the detail drawing.

4.4.2 *Hydrostatic pressure.* Prior to the application of protective finish, the outside, telescoping and inside tube shall be mounted in the device and subjected to the internal hydrostatic pressure. The device used shall internally seal the tube within 1½ inches from each end without applying any axial load other than that applied by the liquid pressure.

4.4.3 *Shear pin strength.* After application of protective finish, the pin shall be subjected to the shear pin strength test in accordance with the requirements of 3.6. The test shall be performed in the fixture in conjunction with the tensile testing machine.

4.4.4 *Structural strength.* The catapult shall be placed in the tensile testing machine and subjected to the specified tension and compression loads for determination of compliance with 3.7. The test shall be performed as follows:

(a) Apply a preload of 500 pounds in tension and adjust the dial indicator to zero inches.

(b) Apply the load in tension so as to move the mounting points away from each other.

(c) Release the load in tension to 500 pounds and record the dial indicator reading.

Repeat steps b and c.

(d) Apply a preload of 500 pounds in compression and adjust the dial indicator to zero inches.

(e) Apply the load in compression so as to move the mounting points toward each other.

(f) Release the compression load to 500 pounds and record the dial indicator reading.

Repeat steps e and f.

4.4.5 *Breakaway torque.* A torque wrench set at the torque requirements specified in 3.8.1 shall be used to determine the adequacy of the joints. The assembly shall be secured in the fixture prior to application of the torque.

4.4.6 *Protective finish.* Test procedures for protective finish shall be in accordance with the applicable Military Standard referenced on the drawing.

4.4.7 *Residual magnetism.* The metal parts assembly shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the assembly 6 inches from and at the same height as the compass in the north-south horizontal position (trunnion end at south) with the compass set in an east-west heading as shown in figure 1. The assembly shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 180 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass.

5. PREPRATION FOR DELIVERY

5.1 Packing and marking. Packing and marking shall be in accordance with Drawings F8797431 and F8797432.

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5.2 Data cards. Data card information shall be as specified in MIL–A–2550.

6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

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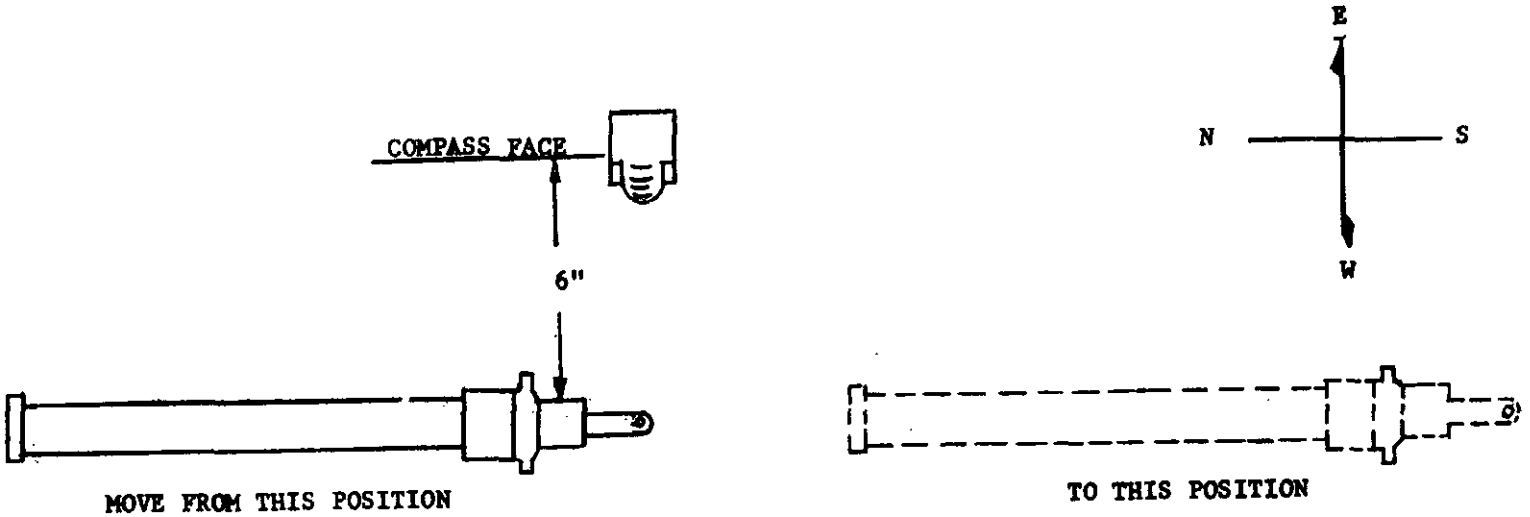
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PILOT'S STANDBY COMPASS, TYPE AN-5766



PLAN VIEW

NOTE:

CATAPULT AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

LAYOUT FOR DETERMINATION OF RESIDUAL MAGNETISM IN CATAPULT, AIRCRAFT EJECTION SEAT, M3A1

FIGURE 1

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